

Nanyang Technological University
SPMS/Division of Mathematical Sciences

2020/2021 Semester 1

MH2814 Probability & Statistics

Test 1 (Mock)

Name &
Matriculation Number

Score:

Mock Test

Answers

.....

1. (30 Points) What is the probability of getting a total of 6 or 10 when a pair of fair dices is tossed?

Answer: We have two cases:

- There are 5 ways to get the total as 6:

$$(1, 5), (5, 1), (2, 4), (4, 2), (3, 3)$$

and the probability for the total of 6 is

$$\frac{1}{6} \times \frac{1}{6} \times 5 = \frac{5}{36}.$$

- There are 3 ways to get the total as 10:

$$(4, 6), (6, 4), (5, 5)$$

and the probability for the total of 10 is

$$\frac{1}{6} \times \frac{1}{6} \times 3 = \frac{5}{36}.$$

So, by the additive rule, the probability of getting a total of 6 or 10 is

$$\frac{5}{36} + \frac{3}{36} = \frac{8}{36} = \frac{2}{9}.$$

2. (40 Points) Suppose that an urn contains 8 red balls and 4 white balls. We draw 2 balls from the urn without replacement.

- (a) If we assume that at each draw each ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red?
- (b) Now suppose that the balls have different weights, with each red ball having weight r and each white ball having weight w . Suppose that the probability that a given ball in the urn is the next one selected is its weight divided by the sum of the weights of all balls currently in the urn. Now what is the probability that both balls are red?

Answer: Let R_1 and R_2 denote, respectively, the events that the first and second balls drawn are red. Now, given that the first ball selected is red, there are 7 remaining red balls and 4 white balls, so

$$P(R_2|R_1) = 7/11.$$

As $P(R_1)$ is clearly $8/12$, the desired probability is

$$P(R_1R_2) = P(R_1)P(R_2|R_1) = \left(\frac{8}{12}\right)\left(\frac{7}{11}\right) = \frac{14}{33}.$$

For part (b), we again let R_i be the event that the i -th ball chosen is red and use $P(R_1R_2) = P(R_1)P(R_2|R_1)$. Now, number the red balls, and let B_i , $i = 1, \dots, 8$, be the event that the first ball drawn is red ball number i . Then

$$P(R_1) = P\left(\bigcup_{i=1}^8 B_i\right) = \sum_{i=1}^8 P(B_i) = 8 \frac{r}{8r + 4w}.$$

Moreover, given that the first ball is red, the urn then contains 7 red and 4 white balls. Thus,

$$P(R_2|R_1) = \frac{7r}{7r + 4w}.$$

Hence, the probability that both balls are red is

$$P(R_1R_2) = \frac{8r}{8r + 4w} \times \frac{7r}{7r + 4w}.$$

3. (30 Points) Let X be a Poisson random variable with parameter λ . Prove that

$$\mathbb{E}(X^n) = \lambda \mathbb{E}((X+1)^{n-1}).$$

Now use this result to show that $\mathbb{E}[X^3] = \lambda^3 + 3\lambda^2 + \lambda$.

Proof:

$$\begin{aligned}\mathbb{E}(X^n) &= \sum_{k=1}^{\infty} k^n e^{-\lambda} \frac{\lambda^k}{k!} \\ &= \sum_{k=0}^{\infty} (k+1)^n e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!} \\ &= \lambda \sum_{k=0}^{\infty} (k+1)^{n-1} e^{-\lambda} \frac{\lambda^k}{k!} \\ &= \lambda \mathbb{E}[(X+1)^{n-1}].\end{aligned}$$

Therefore

$$\mathbb{E}(X^3) = \lambda \mathbb{E}((X+1)^2) = \lambda \mathbb{E}(X^2 + 2X + 1) = \lambda(\lambda(\lambda+1) + 2\lambda + 1) = \lambda^3 + 3\lambda^2 + \lambda.$$